

Speaker Notes For Presentation

Presentation Tips for these Slides (for a 30-40 min talk):



- **Slide 3 (Intro):** Spend about 3-4 minutes here. Paint a picture for the panelists. Give a real-world example:
"Imagine a power grid operator. An AI tells them the grid is failing, but doesn't tell them *why*. If they act blindly, they could shut down power to a hospital unnecessarily."
- **Slide 4 (The Gap):** Spend another 4 minutes here. This is the **most important slide for setting up your project**. You need to convince the panel that current math-based XAI (like SHAP) is fundamentally broken for physical engineering because it doesn't understand "weather" or "network lag."

Speaker Notes & Tips for these Slides



- **Slide 5 (Objectives):** Highlight the word "**Offline**".
Data privacy is a massive concern in critical infrastructure.
Emphasizing that your LLM runs locally (Ollama) shows you understand real-world engineering constraints.
- **Slide 6 (Metrics):** *Spend 4-5 minutes here.*
This is where you directly quote your Q&A document. Say aloud:
"Standard metrics like AUC-ROC are irrelevant for explanation quality. If the power grid is failing, a SHAP score doesn't tell the operator to activate the backup transformer. Actionability does."

[Transition from previous slide]

"We just discussed why Trust, Reasonableness, and Actionability are our core metrics. But to truly understand *why* these metrics matter, we have to look at the human ecosystem inside a Cyber-Physical System. Explanations are not 'one-size-fits-all.'"

[Click to this Slide - Pause for a second so they can see the diagram]

"What you see here is a Use Case workflow demonstrating how XAI actually operates in a live industrial environment. Notice the diverse range of actors on the perimeter. We don't just have one 'user.' We have XAI System Operators, Expert Engineers, Internal Auditors, and Repair Technicians.

In a traditional 'Black Box' system, the AI simply outputs a prediction—like 'System Failure in 10 minutes'—and leaves all of these stakeholders completely blind.

But in a 'Glass Box' ecosystem, look at the interactions happening in the center:

- The **System Operator** (on the left) is querying the AI for *local, instance-level explanations* so they can make immediate, high-stakes decisions under time pressure.
- Meanwhile, the **Expert User** is extracting *feature importance* and learning new decision rules to update the system's logic.
- On the right side, the **Internal Auditor** relies on the XAI to generate compliance reports and check for systemic bias.

- And the **Repair Technician** uses the explained Remaining Useful Life (RUL) to schedule physical maintenance without wasting resources.

[The "So What?" / Transition to the next slide]

"The key takeaway from this workflow is that **Explainable AI is the central nervous system of human-machine collaboration**. If the explanations lack context, the Operator makes the wrong decision, the Technician orders the wrong parts, and the entire ecosystem breaks down.

This exact workflow is what inspired the design of our Multi-Agent Architecture, which I will walk you through next..."

- **Slide 7 (Detection):** You put the "Invigilator Defense" right on the slide.
When you get to that box, you can smile and say, *"We are often asked why we didn't use a Neural Network to detect the anomaly..."* and then answer it perfectly. It shows extreme confidence.
- **Slide 8 (Multi-Agent):** Walk them through the "debate" happening in the code.
Agent 1a is the old way, Agent 1b is the new way, and Agent 2 acts as the judge. It makes your complex code sound like a very logical, structured courtroom.

 Speaker Notes & Tips for the Final Slides:



- **Slide 9 (Dashboard):** Keep this brief.
Point to the image and say, *"This is the Streamlit UI the facility manager actually sees. Notice how we save computing power by only triggering the LLMs when the red anomaly spikes occur."*
- **Slide 10 (Comparison):** This is your **"Aha!" moment**.
You want to guide their eyes to the difference between the two images. Say: *"On the left, the traditional model wastes time and money by sending a repair crew to fix a pump that isn't actually broken. On the right, our model understands the weather and network lag, preventing a false alarm."*
- **Slide 11 (Scorecards):** Remind them of the "Evaluation Bottleneck."
Say: *"Instead of waiting weeks to survey human engineers, our third Agent instantly scores these explanations, proving mathematically that our framework builds more operator trust."*
- **Slide 12 (Conclusion/Future Work):** End with high energy.
The panel loves students who look beyond their current code. Mentioning "EEG scans" and "haptic feedback smart-gloves" for future work shows you are thinking at a Master's/Ph.D. level.



Speaker Script (How to present these slides)

Slide 7: The Data Challenge (Why Simulate?)

"Before we look at the architecture, we have to address the elephant in the room: the data. A common question we get is why we didn't just download a dataset from Kaggle or ask a local power plant for their data.

There are two major reasons. First, Power Grids and Water Plants are critical national infrastructure. Their SCADA systems are air-gapped and highly classified to prevent cyberattacks—they do not share raw telemetry.

Second, and more importantly for our research, real datasets only record the 'What'—a pump broke, or voltage dropped. They don't synchronize with the 'Why'—like a lightning storm or network latency. To prove our Context-Aware

framework actually works, we needed a dataset that perfectly captured both the internal sensors AND the external environment."

Slide 8: Dataset Specifications

"To solve this, we mathematically synthesized our own telemetry. We built two distinct 1,000-sample datasets representing 15-minute operational intervals. We ensured this data was highly legitimate by anchoring it to real engineering standards using Gaussian distributions. For example, our Power Grid idles normally at a strict 50 Hertz frequency and 132 kilovolts, while our Water Plant idles at 50 psi.

We then injected two specific types of anomalies: true Mechanical Failures, where equipment just breaks, and Contextual Anomalies, where a severe Heatwave or Storm causes the sensors to fluctuate. This allowed us to definitively test if our AI Agents could tell the difference between a broken machine and a machine fighting bad weather."

Slide 1: Title Page



Script:

"Good morning, esteemed members of the panel. I am Subhranshu Panda, and along with my partner Shreyansh Gupta, we are honored to present our final year project: **'Building the Glass Box: A Human-Centered Framework for Explainable AI in Cyber-Physical Systems.'** We worked on this research under the invaluable guidance of Prof. Bharati Mishra. Our project focuses on making AI decision-making transparent and trustworthy for the critical infrastructure that runs our world."

Slide 2: Presentation Outline



Script:

"Today, we will walk you through our journey. We'll start by defining the 'Black Box' crisis in current systems and identifying the 'Context-Awareness Gap' in existing AI. Then, we will detail our proposed Multi-Agent architecture, show you our live software prototype, and finally, present a novel way to evaluate AI explanations using an 'LLM-as-a-Judge' approach."

Slide 3: The Rise of AI in Cyber-Physical Systems (CPS)



Script:

"To understand our work, we first need to look at Cyber-Physical Systems, or CPS. These are systems where computer algorithms control physical processes—think of smart power grids or water treatment plants.

As we move toward **Industry 5.0**, the goal is no longer just 'automation,' but 'collaboration' between humans and machines. However, there is a major roadblock: **The Black Box Crisis**. High-performance AI models are incredibly accurate, but they are opaque. In a safety-critical environment, if an AI flags a catastrophic anomaly, an operator can't simply take its word for it. They need to know *why* that decision was made before they hit the emergency stop button."

Slide 4: The Crisis of Traditional Explainable AI (XAI)



Script:

"To solve this, the field of Explainable AI, or XAI, was born. You might be familiar with industry standards like SHAP or LIME. But here is the problem: these methods have a '**Context-Awareness Gap**.'

Traditional XAI looks at data in a vacuum. For example, if a water pump vibrations spike, the AI will mathematically flag the 'vibration variable.' But it ignores the world *outside* the sensor. It doesn't know there is a massive storm outside or a heatwave affecting the grid. This leads to false positives and misdiagnoses, which quickly destroys the operator's trust in the system. Our project aims to bridge this gap by bringing 'Context' into the explanation."

Slide 5: Project Objectives & The "Glass Box" Approach



Script:

"Our primary objective was to move from theory to a functional, enterprise-ready software prototype."

1. We built a **Context-Aware Reasoning Engine** that connects internal sensors with external variables like weather and network lag.
2. We prioritized **Cybersecurity-by-Design**, creating a 100% offline pipeline so sensitive data never leaves the local facility.
3. We proved **Scalability** by testing across two different domains: Water and Power.
4. And crucially, we pioneered an **Automated Evaluation** pipeline to score our AI without needing months of manual human surveys.

Slide 6: Redefining Metrics (Trust, Reasonableness, Actionability)

The Deep Concept:

In computer science, we are trained to worship "Accuracy," "Precision," and "F1-Scores." But in a **Cyber-Physical System (CPS)**, the AI isn't just classifying photos of cats and dogs—it is controlling things that can explode, flood, or blackout a city. If an AI is 99% accurate but explains its decision in a way the human operator doesn't understand, the operator will ignore it. You are shifting the focus from *Algorithmic Fidelity* (how true the explanation is to the math) to *Human Utility* (how useful the explanation is to the person pressing the buttons).

Detailed Breakdown:

- **Trust:** This is the baseline. If an operator looks out the window and sees a hurricane, but the AI says "I am 100% sure the pump is vibrating because of a loose screw," the operator immediately loses faith in the system. Trust means the AI's explanation matches the operator's perception of reality.
- **Reasonableness:** This tests whether the AI understands *physics and domain logic*. If voltage drops, a reasonable explanation connects it to high thermal load (heatwave) or a broken line. An unreasonable explanation connects it to something irrelevant.
- **Actionability:** This is the ultimate goal. A bad explanation says: "*Feature 1 (Pressure) is low.*" A good explanation says: "*Network latency is causing sensor lag. Switch to manual override.*"  **Anticipated Panelist Question & Your Defense:**
- **Question:** "*Why didn't you just use standard XAI metrics like SHAP or LIME?*"
- **Your Defense:** "*Because SHAP and LIME output numerical feature weights. If a power grid is failing, handing an operator a chart that says 'Voltage has a SHAP value of 0.85' does not tell them what to do. Our metrics ensure the AI provides a human-readable, causally-grounded instruction, which is what is actually required in a high-stakes crisis.*"

Slide 7: The Data Challenge (Why Simulate?)

The Deep Concept:

In academia, synthesizing (faking) data is sometimes seen as taking the easy way out. You must aggressively frame this as a **forced engineering necessity**. You

couldn't use real data because the exact data you needed *does not exist* in the public domain.

Detailed Breakdown:

- **National Security & Air-Gapping:** SCADA (Supervisory Control and Data Acquisition) systems run water plants and power grids. Because of threats like cyber-terrorism (e.g., the Stuxnet virus that attacked nuclear centrifuges), these systems are "air-gapped"—meaning they are physically disconnected from the internet. They do not publish their raw telemetry to Kaggle or GitHub.
- **The Contextual Labeling Deficit:** This is your strongest argument. Even if a power plant gave you their data, it would only be a spreadsheet of voltages and currents. It would *not* include a column that says "By the way, there was a heatwave at this exact minute." To train and test an AI on **Context-Awareness**, you absolutely require datasets where external events (weather) are perfectly synchronized with internal sensors (voltage).

Anticipated Panelist Question & Your Defense:

- **Question:** *"Isn't your AI just performing well because you created the data yourself?"*
- **Your Defense:** *"We synthesized the data not to make it easier, but to isolate the variables. Real-world data conflates mechanical failures with environmental noise. By mathematically simulating the data, we created controlled scenarios to definitively prove whether the AI could distinguish between a broken machine and a machine reacting to bad weather. We couldn't test that hypothesis without perfectly labeled context."*

Slide 8: Dual-Domain Dataset Synthesis & Validation

The Deep Concept:

This slide proves your synthetic data isn't just random numbers. It proves you used real-world engineering constraints to generate it.

Detailed Breakdown:

- **Gaussian Distributions:** You used statistics (bell curves) to define "normal." For example, a power grid in India operates at 50Hz. It doesn't randomly jump to 45Hz or 55Hz. It hovers tightly between 49.9Hz and 50.1Hz. By forcing your

synthetic data to follow these strict mathematical distributions, it behaves exactly like a real physical sensor.

- **Isolating the Anomalies:** You built two distinct "bad" events.
 1. *Mechanical Fault:* A transformer explodes. The voltage drops immediately.
 2. *Contextual Fault:* A heatwave hits. Everyone turns on their AC units. The voltage drops slowly due to load stress.
The telemetry looks similar, but the *cause* is different. You built this data specifically to test if your AI was smart enough to tell the difference.

Anticipated Panelist Question & Your Defense:

- **Question:** *"How did you decide on the threshold values like 50Hz or 132kV?"*
 - **Your Defense:** *"We anchored our simulation parameters to standard electrical and mechanical engineering operating tolerances. The AI wasn't trained on arbitrary numbers; it was subjected to the same rigid boundary conditions that govern real-world SCADA systems."*
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Slide 9: Operational Workflow (XAI-CPS Ecosystem)

The Deep Concept:

XAI is not "One-Size-Fits-All." What counts as a "good explanation" completely changes depending on who is asking the question. You are showing the panel that you understand enterprise software architecture and the different roles within a real company.

Detailed Breakdown:

- **The System Operator:** They are in the control room. An alarm goes off. They have 60 seconds to act. They need **Local Explainability** (an explanation for *this specific event* right now) that is highly actionable.
- **The Expert Engineer:** They are reviewing the system the next day. They want to know *why* the system failed so they can update the thresholds. They need **Global Explainability** (understanding the AI's overall logic and feature rules).
- **The Auditor:** They don't care about fixing the pump. They care about safety compliance. They need explanations that prove the system isn't biased and is following safety regulations.

- **Your "Glass Box":** Your LLM-based, multi-agent system is powerful because you can change the *persona* of the AI prompt. The same data can be explained simply to the Operator, or highly technically to the Engineer.

Anticipated Panelist Question & Your Defense:

- **Question:** *"If different users get different explanations, doesn't that mean the AI is being inconsistent or hallucinating?"*
- **Your Defense:** *"The underlying causal truth remains exactly the same. What changes is the level of abstraction. Just as a doctor explains a diagnosis differently to a patient than to another surgeon, our framework adapts the complexity of the explanation to match the cognitive load and operational needs of the specific user."*

Slide 10: System Architecture: Step 1 - Anomaly Detection



Script:

"Now, let's look under the hood. Our architecture follows a two-step process.

First is **Rule-Based Detection**. We use hard engineering thresholds because that is the standard in industrial SCADA systems. We don't want the AI to 'guess' if a frequency drop is bad—we know it is. Our contribution isn't the detection; it is the **explanation** that follows."

Slide 11: Architecture Step 2 - Multi-Agent Orchestration



Script:

"The second step is where the magic happens. We used **Microsoft AutoGen** to manage a conversation between three agents, powered by the **Llama 3.2** model running entirely offline via **Ollama**."

- **Agent 1a** simulates the old way (Context-Agnostic).
- **Agent 1b** is our proposed Context-Aware model.
- **Agent 2** is our Expert Evaluator.

By having these agents 'debate' the sensor data and external context, we force the AI to make its reasoning explicit and transparent.

Slide 12: Empirical Results: The "Glass Box" Dashboard



Script:

"We wrapped this entire logic into a **Streamlit Dashboard**. Here you can see the real-time telemetry. Notice that we save significant computing power by only waking up the heavy LLM agents when a red anomaly spike is detected by the rule-based system."

Slide 13: Comparative AI Diagnostics in Action



Script:

"This is the most impactful part of our results.

On the left, the **Traditional AI** sees a voltage drop and immediately tells the operator the transformer is broken. It recommends an expensive repair.

On the right, our **Proposed Framework** looks at the external variables, sees the regional heatwave, and realizes the grid is just overworking. It prevents a false alarm and saves the facility thousands in unnecessary maintenance."

Slide 14: Automated Expert Evaluation (LLM-as-a-Judge)



Script:

"Finally, we solved the 'Evaluation Bottleneck.' Instead of waiting for a human expert to grade these explanations, we used an LLM-as-a-Judge.

We gave our second Agent the persona of a 'Senior Grid Engineer.' It objectively scored both explanations. As you can see, our Context-Aware model consistently scored higher across all metrics because its reasoning was grounded in reality, not just isolated numbers."

Slide 15: Conclusion & Future Research Directions

**Script:**

"In conclusion, we have moved beyond theoretical XAI. We have built a secure, offline, cross-domain prototype that proves context is the key to trust.

Looking ahead, we want to expand this into a decentralized ecosystem of agents and eventually integrate haptic feedback—like smart-gloves—so operators can 'feel' an anomaly explanation in noisy environments. We also plan to validate our AI-judge scores against human eye-tracking data."

Slide 16: Thank You

**Script:**

"That concludes our presentation. We have demonstrated that the 'Glass Box' approach is not only possible but necessary for the future of Industry 5.0. Thank you for your time, and we are now open for questions and discussion."